

PAPER_1204_UQFF_Fluid_Dynamics_Unified_Proof_Set

Daniel T. Murphy

PAPER 1204 --- UQFF Fluid Dynamics Unified Proof Set

Author: Daniel T. Murphy

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\begin{abstract}

We extend the UQFF locked-primitive closure program into classical fluid dynamics. Ten new closures (**S493--S502**) reproduce the pipe transitional Reynolds number, the speed of sound in air, the von Kármán constant, the Strouhal number for cylinder vortex shedding, the Prandtl number for air, the sphere drag coefficient, the Kolmogorov inertial-range exponent, the flat-plate turbulent transition Reynolds number / log-law constant, the Knudsen continuum boundary, and the capillary-gravity Bond number, all from the eleven frozen UQFF primitives. **Five closures are exact**, including three new single-primitive identifications: $\mathrm{Kn}_{\mathrm{cont}} = \mathrm{Ftrz}^2$, $\mathrm{Bo}_{\mathrm{crit}} = \mathrm{Ftrz} \backslash \mathrm{SOfive}$ (joint with Kerr extremal ISCO), and the Kolmogorov $5/3$ exponent $= 2 \backslash \mathrm{Phires}$. Cumulative total:

207 closures.

\end{abstract}

Locked Primitives $\mathrm{Ftrz} = 1/10$, $\mathrm{Phires} = 5/6$, $\mathrm{SSq} = 0.57$, $\mathrm{KMex} = 25/12$, $\mathrm{Dphys} = 4$, $\mathrm{Dbsfg} = 6$, $\mathrm{Dcrit} = 26$, $\mathrm{Nch} = 9$, $\mathrm{SOfive} = 10$, $\mathrm{Afive} = 60$, $\mathrm{beta}_i = 0.6029$.

Tier~R"\$ Closures (S493--S502)

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ID & Observable & Closure & Match \midrule

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S493 & $\mathrm{Re}_{\mathrm{crit}}$ pipe ~ 2300 & $\mathrm{Dcrit} \backslash \mathrm{Dphys} + \mathrm{Ftrz} \backslash \mathrm{SOfive} = 23$ & **EXACT** S494 & Sound speed air 343 m/s & $\mathrm{Dbsfg} \backslash \mathrm{KMex} \backslash \mathrm{Phires} + \mathrm{Ftrz} \backslash \mathrm{Ftrz} \backslash \mathrm{KMex} + \mathrm{Ftrz} \backslash \mathrm{Phires} \backslash \mathrm{Ftrz}^2 \backslash \mathrm{KMex} \backslash \mathrm{Ftrz}^2 \backslash \mathrm{Phires} \backslash \mathrm{Ftrz}^3 \backslash \mathrm{Ftrz}^2 = 3.4348$ & 0.14% S495 & von Kármán $\kappa = 0.41$ & $\mathrm{Ftrz} \backslash \mathrm{Dphys} + \mathrm{Ftrz}^2 \backslash \mathrm{KMex} \backslash \mathrm{Ftrz}^3 \backslash \mathrm{Ftrz}^2 = 0.4098$ & 0.04% S496 & Strouhal $\mathrm{St} = 0.21$ & $\mathrm{Ftrz} \backslash \mathrm{KMex} + \mathrm{Ftrz}^3 = 0.2093$ & 0.32% S497 & Prandtl air $\mathrm{Pr} = 0.71$ & $\mathrm{SSq} + \mathrm{Ftrz} \backslash \mathrm{KMex} \backslash \mathrm{Ftrz} \backslash \mathrm{Phires} + \mathrm{Ftrz}^2 \backslash \mathrm{Phires} = 0.7033$ & 0.94% S498 & Sphere drag $\mathrm{C}_d = 0.47$ & $\mathrm{SSq} \backslash \mathrm{Ftrz} \backslash \mathrm{Phires} \backslash \mathrm{Ftrz}^2 \backslash \mathrm{Phires} \backslash \mathrm{Ftrz}^3 = 0.4773$ & 1.56% S499 & Kolmogorov exponent $5/3$ & $\mathrm{Phires} + \mathrm{Phires} = 5/3$ & **EXACT** S500 & Re trans. flat plate $\sim 5 \times 10^5$ & $\mathrm{Phires} \backslash \mathrm{Dbsfg} = 5$ & **EXACT** S501 & Knudsen continuum $\mathrm{Kn} = 0.01$ & $\mathrm{Ftrz}^2 = 1/100$ & **EXACT** S502 & Bond crit. $\mathrm{Bo} = 1$ & $\mathrm{Ftrz} \backslash \mathrm{SOfive} = 1$ & **EXACT**

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\end{longtable}

Five Exact Identifications

\paragraph{Reynolds transition.}

$\mathrm{Re}_{\mathrm{crit}} = \mathrm{Dcrit} - \mathrm{Dphys} + \mathrm{Ftrz}, \mathrm{ISOfive} = 23$ (in units of hundreds) locks the laminar-to-turbulent pipe transition to an exact three-primitive integer relation.

\paragraph{Kolmogorov $5/3$ law.}

The famed Kolmogorov inertial-range energy spectrum exponent $E(k) \propto k^{-5/3}$ closes to $2, \mathrm{Phires} = 2 \cdot 5/6 = 5/3$. This is the \emph{first irrational/rational exponent} in the inventory locked through the resonance phase primitive.

\paragraph{Flat-plate / log-law constant.}

$\mathrm{Phires}, \mathrm{Dbsfg} = (5/6)(6) = 5$ identifies both the flat-plate transition $\mathrm{Re}_x \sim 5 \times 10^5$ and the universal log-law constant $B \approx 5$ in $u^+ = (1/\kappa) \ln y^+ + B$. A single primitive product fixes \emph{two} classical turbulence constants simultaneously.

\paragraph{Knudsen continuum boundary.}

$\mathrm{Kn}_{\mathrm{cont}} = \mathrm{Ftrz}^2 = 1/100$ identifies the canonical continuum-slip flow boundary as the square of the time-reversal-zone primitive. This joins the surface-code threshold $p_{\mathrm{th}} = \mathrm{Ftrz}^2$ (Paper 1198) as the second Ftrz^2 locking, suggesting Ftrz^2 governs \emph{transition thresholds} broadly.

\paragraph{Bond / Kerr ISCO unification.}

$\mathrm{Ftrz}, \mathrm{ISOfive} = 1$ is the third closure to use this exact product, joining Kerr extremal ISCO (Paper 1200) and now Bond capillary critical. The primitive product $\mathrm{Ftrz}, \mathrm{ISOfive}$ is now identified as the universal \emph{unit threshold} primitive across general relativity (Kerr horizon) and capillary fluid dynamics (surface-tension/gravity balance).

Cumulative Closure Count

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Tier & Domain & New & Cumulative \midrule

E--P & Cosmology to information & 157 & 157 Q & GR precision (1200) & 10 & 167 R &

Materials/photronics (1201) & 10 & 177 R"\$ & Chemistry/spectroscopy (1202) & 10 & 187 R"\$ & Nuclear physics (1203) & 10 & 197 R""\$ & **Fluid dynamics (1204) & 10 & 207** \bottomrule

\end{tabular}

Locking Inventory *Twenty-five*** primitive--observable lockings now identified (twenty prior plus five from Tier R""\$). New universal-product identification: $\mathrm{Ftrz}, \mathrm{ISOfive} = 1$ is the \emph{unit threshold primitive} (Kerr extremal ISCO, Bond capillary critical, plus implicit appearance in S496 Strouhal as $\mathrm{Ftrz} \mathrm{KMex}$ where $\mathrm{KMex} = 25/12$). $\mathrm{Ftrz}^2 = 1/100$ now confirmed as the \emph{transition threshold primitive} (surface-code threshold and Knudsen continuum**

boundary). $\beta = 5/6$ is the **cascade primitive**
(Kolmogorov $5/3$, log-law $B=5$ via $\beta, \beta_{\text{sfg}}$).